Healthcare/IoT Cybersecurity Testbed: Milestone 3

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Milestone Completion

Task	Completion %	Justin	Nathan	Ipule	To-Do
Tooling Demos	75%	50%	25%	0%	Finish current demos and make plans for demos of additional functionality
Device Examination	60%	50%	10%	0%	Further examine device chipsets, interfaces, and documentation.
Tooling implementation	66%	33%	33%	0%	Continue integrating analysis tools with backend services.
Primary Scripting Round	100%	100%	0%	0%	Begin secondary scripting round and integrations
Website/AWS Infrastructure development	75%	0%	50%	25%	Fix frontend build issues in the GitHub Actions deployment pipeline.

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Cloud Infrastructure & Deployment Review

- **Terraform Infrastructure-as-Code -** Migrated AWS infrastructure to Terraform, enabling reproducible cloud deployments and faster backend development
  - This allows for more consistent and faster updates to the backend
  - It also allows AWS resources to match be current code base in real time with no specific actions needed

- **Automated CI/CD Pipeline**

  Implemented GitHub Actions to automate infrastructure updates and application deployment to AWS.

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Platform Integration Review

Frontend + Cloud Integration - Deployed the web interface and tested integration with AWS services, authentication, and containerized analysis tools.

Security & Authentication - Integrated AWS Cognito to provide secure user authentication and identity management for the platform.

Containers & Backend - Improved compatibility with lambda functions and added more options and tested to confirm they work with uploaded containers

FRONTEND ARCHITECTURE

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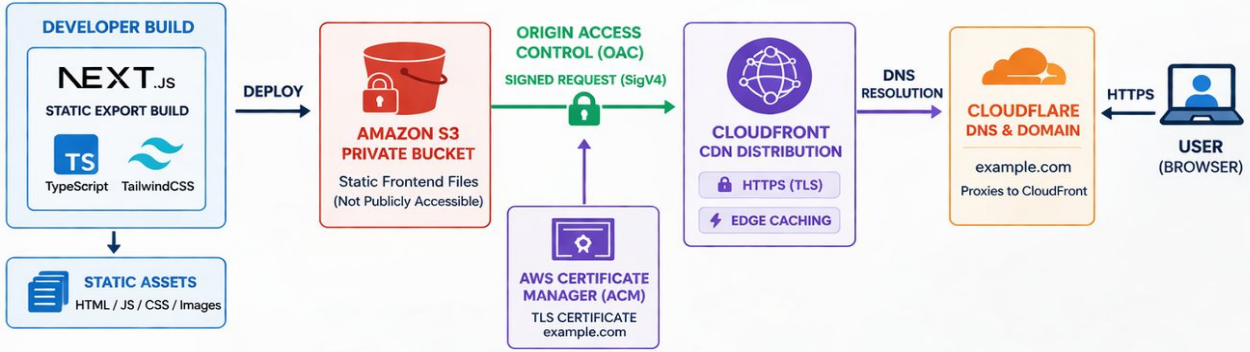
- Built **Next.js** web interface using **TypeScript** and **TailwindCSS.**
- Configured **static export build** to remove server dependency and enable CDN hosting.
- Implemented **responsive dashboard UI** for device vulnerability monitoring.

SECURITY & TLS

- Configured **private S3 bucket** with **CloudFront access control (OAC).**
- Ensured frontend assets are served **securely via CDN.**
- Set up **AWS Certificate Manager (ACM) TLS certificate** for HTTPS.
- Enabled **secure HTTPS delivery** through CloudFront.

DOMAIN & NETWORKING

- Integrated **Cloudflare domain** with the **CloudFront distribution.**
- Configured **DNS routing** from **Cloudflare** to AWS infrastructure.



SECURE, SCALABLE, CDN-FIRST FRONTEND DELIVERY

Static Next.js application hosted on a **private S3 bucket**, delivered globally through CloudFront CDN with full HTTPS encryption and Cloudflare-managed domain routing.

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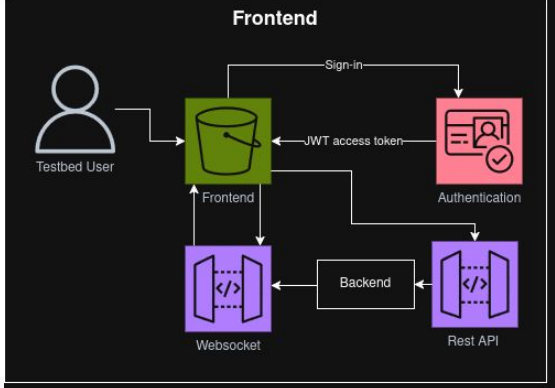

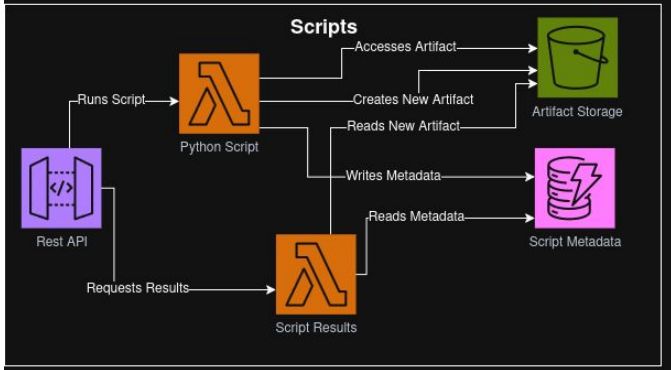

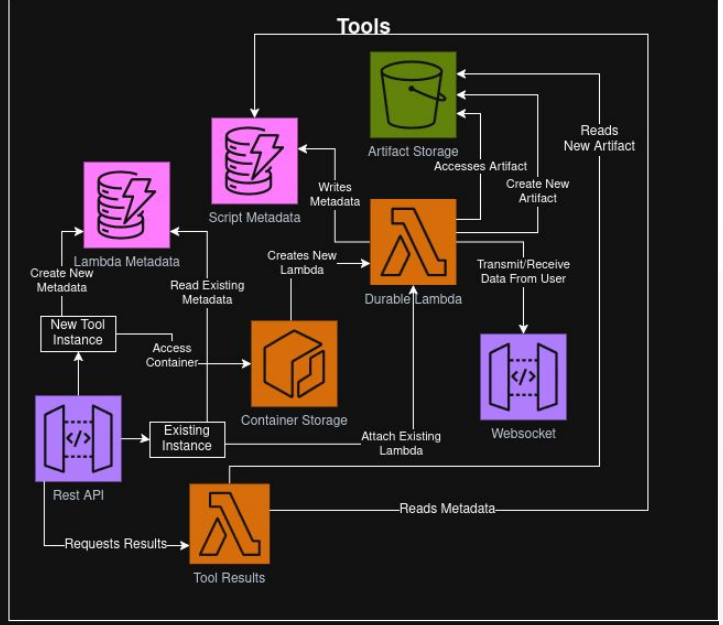

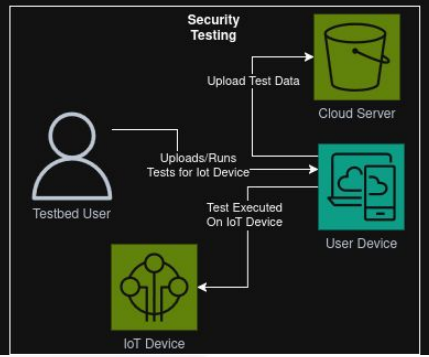

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Scripting, RE, and Demo Review from Milestone 2

- Completed additional research into IoT device exploitation in preparation for device examination in the IoT lab
- Bluetooth is the most common interface held in common by all devices, which was a type of RE that we had never tackled before
- Chipset documentation will be important for low level tasks and allowed manufacturers to utilize the test suite to test attack vectors will require a chipset list
- Existing scripts are being turned into demos, already integrated into AWS, which has been integrated with the website
- These preliminary demos will serve as the proof-of-concept for replicable and possibly automated static analysis based exploits

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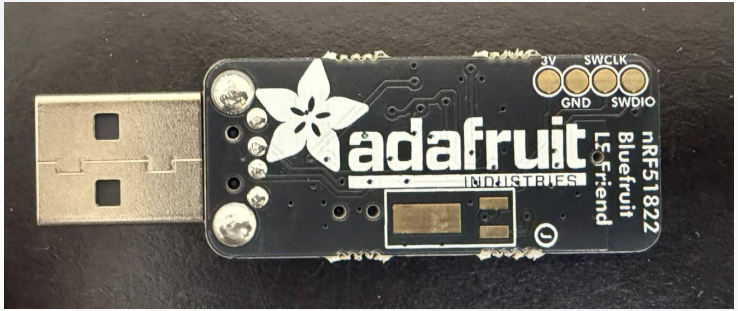
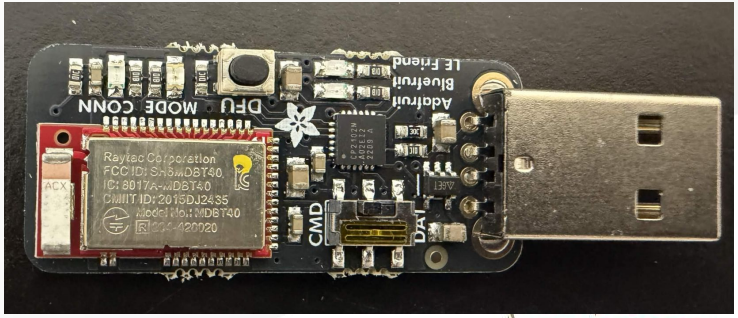
Script, RE, and Demo Progress for Milestone 3

- Bluetooth sniffer is secured and undergoing testing (Adafruit nRF51822 Bluefruit LE)
- Examined oximeters and smart devices, which will require more tools to reverse via disassembly
- Additional scripting for bluetooth exploitation is completed and ready for testing





Devices In-progress of RE

Script, RE, and Demo Progress for Milestone 3

- Disassembled multiple network adapters, with no UART/JTAG pinouts found
- Identified wifi/bluetooth chipsets for network adapters and documented
- Knowledge of chipsets will allow for exploiting documented vulnerabilities in the demos
- Potential for integrations with the CVE database in the documentation section of the test suite
- Discussed additional tooling with Dr. Sudhakaran including (EEPROM, On-board chip reader, heat gun, and soldering iron)

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Documented Chipsets/PCBs

Network Adapter

Panda Wireless PAU0B -> Mediatek MT7610UN 2234-AMAL FTPBS804 Chipset

Alfa Networks AWUSO36ACH AC1200

-> Realtek RTL8812AU 08MA1G2 G034 Chipset

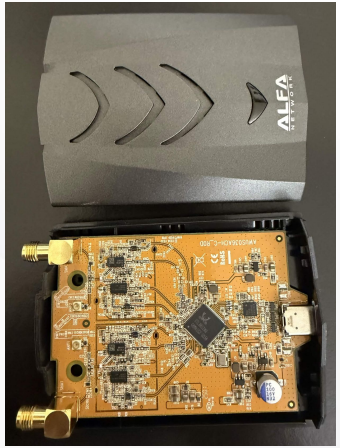
Alfa Networks AWUS036ACM

-> MediaTek MT7612UN 2436 BMJL

BTPB7A08 Chipset

Alfa Networks AWUS036ACS

-> RealTek RTL8811AU Chipset



Current Script Review

File Utilities

analyze_entropy.py - uses binwalk to examine file entropy

lambda_invoke.py - assists in lambda code execution

parse_firmware.py - parses firmware files over ssh or http protocols

analyze_bluetooth_pcap.py - perform preliminary analysis of pcap files

sniffer.py - script with accompanying API for automatically ripping pcap files over bluetooth using wireshark

PyGhidra Utilities

unsanitized_text_entries_scanner.py - scans files for poor text entry handling for potential ret2* exploitations

buffer_overflow_scanner.py - scans for potential buffer overflow exploits

rop_gadgets_scanner.py - scans for ROP gadgets for ropper exploitation

Milestone 4 Plans

Task	Justin	Nathan	Ipule
Tooling Demos	Continue developing PCAP/firmware demo workflows	Support backend integration for demo data	Integrate demo results into web interface
Device Examination	Continue testing IoT devices and data extraction methods	Assist with infrastructure support if needed	Assist with examination
Tooling implementation	Improve analysis tools and workflows	Integrate tools with AWS backend services	Connect tools to frontend for user interaction

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Secondary Scripting Round and Integrations	Develop and optimize analysis scripts	Support execution environment for scripts	Integrate script outputs into UI
Infrastructure development	Assist with integration if needed	Expand infrastructure for S3 storage and processing pipelines	Assist with integration between frontend and backend
Web Interface Development	Assist with development if needed	Support API/backend endpoints	Implement firmware upload functionality and connect to S3

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Thanks for listening!
Questions?

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